

## Binary Conversion

### Learning Objectives

- Convert decimal numbers to binary
- Convert binary numbers to decimal
- Verify conversions using the doubling method

### Engage (5-7 minutes)

Think of measuring land in acres versus hectares—same area, different units. Binary and decimal are like that for numbers. Today we learn to convert between them like translating between two languages.

### Explore (10-12 minutes)

Use the division method: divide 25 by 2 repeatedly and write remainders. Read remainders bottom-up to get binary. Then use the addition method: convert 1101 back to decimal by adding  $8+4+0+1$ . Verify both give the same answer.

### Explain (10-12 minutes)

To convert decimal to binary: divide by 2, record remainder, repeat until quotient is 0. Read remainders from bottom to top. To convert binary to decimal: multiply each digit by its position value (1, 2, 4, 8, 16...) and add results. Always verify by converting back.

### Elaborate (8-10 minutes)

Convert these numbers: decimal 30 to binary, binary 10110 to decimal. Then try a challenge—find the binary for your age. Use the verification method to check each answer is correct.

### Evaluate (5-8 minutes)

Exit ticket: Convert decimal 19 to binary and binary 1100 to decimal. Show one conversion step-by-step.